

Q1

```

241-0583 Lab...d Raza (55)
1  -- Q1. AFTER INSERT: Captain must belong to same team as player
2  CREATE TRIGGER trg_player_captain_team
3  ON player
4  AFTER INSERT
5  AS
6  BEGIN
7      IF EXISTS (
8          SELECT 1 FROM inserted i
9          JOIN player p ON i.captain_id = p.pid
10         WHERE i.captain_id IS NOT NULL
11             AND i.tid != p.tid
12     )
13     BEGIN
14         RAISERROR('Captain must belong to the same team.', 16, 1);
15         ROLLBACK;
16     END
17 END;
18 GO
19
20 -- Q2. AFTER UPDATE: Cannot change tid if player is someone else's captain
21 CREATE TRIGGER trg_player_tid_update
22 ON player

```

110 % [Error Icon] [Warning Icon] [Info Icon] [Up Arrow] [Down Arrow]

Ln: 1, Ch: 1 (431 chars, 18 lines) SPC CRLF UTF

**Messages**

Commands completed successfully.

Completion time: 2026-04-17T12:32:06.4114297-07:00

```
241-0583 Lab...d Raza (78)  >
1  -- Q1
2  CREATE TRIGGER trg_player_captain_team
3  ON player
4  AFTER INSERT
5  AS
6  BEGIN
7      IF EXISTS (
8          SELECT 1 FROM inserted i
9          JOIN player p ON i.captain_id = p.pid
10         WHERE i.captain_id IS NOT NULL
11             AND i.tid != p.tid
12     )
13     BEGIN
14         RAISERROR('Captain must belong to the same team.', 16, 1);
15         ROLLBACK;
16     END
17 END;
18
19 -- captain_id 101 belongs to team 1, but new player is in team 2 -> trigger fires
20 INSERT INTO player VALUES ('Test Player', 206, 25, 'Batsman', 60000, 2, 101, 25, 2);
21
22 -- Q2
23
```

110 % ▾

Messages

A new player has been added.  
Msg 50000, Level 16, State 1, Procedure trg\_player\_captain\_team, Line 14 [Batch Start Line 18]  
Captain must belong to the same team.  
Msg 3609, Level 16, State 1, Procedure trg\_insteadof\_insert\_player, Line 10 [Batch Start Line 18]  
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-04-17T12:44:08.9500569-07:00

Ln: 19, Cln: 1 (164 chars, 2 lines) SPC | CF

## Q2

```

24L-0583 Lab...d Raza (78)
19 -- captain_id 101 belongs to team 1, but new player is in team 2 → trigger fires
20 INSERT INTO player VALUES ('Test Player', 206, 25, 'Batsman', 60000, 2, 101,25,2);
21
22 -- Q2
23 CREATE TRIGGER trg_player_tid_update
24 ON player
25 AFTER UPDATE
26 AS
27 BEGIN
28     IF EXISTS (
29         SELECT 1 FROM inserted i
30         JOIN deleted d ON i.pid = d.pid
31         JOIN player p ON p.captain_id = i.pid
32         WHERE i.tid != d.tid
33     )
34     BEGIN
35         RAISERROR('Cannot change team: player is assigned as captain of other players.', 16, 1);
36         ROLLBACK;
37     END
38 END;
39
40 -- pid 101 is captain of pid 102, moving him to team 2 → trigger fires
41 UPDATE player SET tid = 2 WHERE pid = 101;

```

110 % 2 0 ↑ ↓ Ln: 22, Ch: 1 (398)

Messages

Commands completed successfully.

Completion time: 2026-04-17T12:44:33.9879080-07:00

```

24L-0583 Lab...d Raza (78)
22 -- Q2
23 CREATE TRIGGER trg_player_tid_update
24 ON player
25 AFTER UPDATE
26 AS
27 BEGIN
28     IF EXISTS (
29         SELECT 1 FROM inserted i
30         JOIN deleted d ON i.pid = d.pid
31         JOIN player p ON p.captain_id = i.pid
32         WHERE i.tid != d.tid
33     )
34     BEGIN
35         RAISERROR('Cannot change team: player is assigned as captain of other players.', 16, 1);
36         ROLLBACK;
37     END
38 END;
39
40 -- pid 101 is captain of pid 102, moving him to team 2 → trigger fires
41 UPDATE player SET tid = 2 WHERE pid = 101;
42
43 -- Q3

```

110 % 2 0 ↑ ↓ Ln: 40, Ch: 1 (114 chars, 2 lines) SPC

Messages

(0 rows affected)

Msg 50000, Level 16, State 1, Procedure trg\_player\_tid\_update, Line 14 [Batch Start Line 39]

Cannot change team: player is assigned as captain of other players.

Msg 3609, Level 16, State 1, Line 41

The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-04-17T12:44:48.4828339-07:00

## Q3

```
241-0583 Lab...d Raza (78)
37 END
38 END;
39
40 -- pid 101 is captain of pid 102, moving him to team 2 + trigger fires
41 UPDATE player SET tid = 2 WHERE pid = 101;
42
43 -- Q3
44 CREATE TRIGGER trg_player_delete_reassign
45 ON player
46 INSTEAD OF DELETE
47 AS
48 BEGIN
49     UPDATE player
50     SET captain_id = NULL
51     WHERE captain_id IN (SELECT pid FROM deleted);
52
53     DELETE FROM player
54     WHERE pid IN (SELECT pid FROM deleted);
55 END;
56
57 DELETE FROM player WHERE pid = 101;
58 SELECT pid, pname, captain_id FROM player WHERE pid = 102;
59
60 -- pid 101 is captain of pid 102, deleting 101 + 102's captain_id becomes NULL first, then 101 is deleted
61 DELETE FROM player WHERE pid = 101;
62 SELECT pid, pname, captain_id FROM player WHERE pid = 102;
63
```

91% 2 0 1 Lnt: 43, Ch: 1 (363 chars, 16 lines) SPC CRLF U

Messages

Command completed successfully.

Completion time: 2024-04-17T19:16:03.2427246-07:00

```
SQLQuery3.sql Raza (83)* 241-0583 Lab...d Raza (78)
40 -- pid 101 is captain of pid 102, moving him to team 2 + trigger fires
41 UPDATE player SET tid = 2 WHERE pid = 101;
42
43 -- Q3
44 CREATE TRIGGER trg_player_delete_reassign
45 ON player
46 INSTEAD OF DELETE
47 AS
48 BEGIN
49     UPDATE player
50     SET captain_id = NULL
51     WHERE captain_id IN (SELECT pid FROM deleted);
52
53     DELETE FROM player
54     WHERE pid IN (SELECT pid FROM deleted);
55 END;
56
57 -- FK Constraint was not letting us delet from player, because pid 101 is also in the performance table.
58 ALTER TABLE performance
59 DROP CONSTRAINT FK_performance_pid_SAE8289;
60 DELETE FROM player WHERE pid = 101;
61 SELECT pid, pname, captain_id FROM player WHERE pid = 102;
62
63 -- pid 101 is captain of pid 102, deleting 101 + 102's captain_id becomes NULL first, then 101 is deleted
64 DELETE FROM player WHERE pid = 101;
65 SELECT pid, pname, captain_id FROM player WHERE pid = 102;
66
```

91% 2 0 1 Lnt: 57, Ch: 1 (480 chars, 9 lines) SPC CRLF U

Results

| pid | pname | captain_id    |
|-----|-------|---------------|
| 1   | 102   | Shaheen Afidi |

Messages

| pid | pname | captain_id    |
|-----|-------|---------------|
| 1   | 102   | Shaheen Afidi |

Q4

```
241-0583 Lab...d Raza (78)
58 DELETE FROM player WHERE pid = 101;
59 SELECT pid, pname, captain_id FROM player WHERE pid = 102;
60
61 -- Q4
62 CREATE TRIGGER trg_match_same_team
63 ON match
64 AFTER INSERT
65 AS
66 BEGIN
67     IF EXISTS (
68         SELECT 1 FROM inserted
69         WHERE team1_id = team2_id
70     )
71     BEGIN
72         RAISERROR('A team cannot play against itself.', 16, 1);
73         ROLLBACK;
74     END
75 END;
76
77 -- team1_id = team2_id = 1 + trigger fires
78 INSERT INTO match VALUES (301, '2025-01-01', 'Lahore', 1, 1, NULL, 1);
79
```

110 % 2 0 11 Ln: 75, Ch: 5 (277 chars, 15 lines) SPC CRLF

Messages

Commands completed successfully.

Completion time: 2026-04-17T12:46:05.3566218-07:00

```
241-0583 Lab...d Raza (78)
64 AFTER INSERT
65 AS
66 BEGIN
67     IF EXISTS (
68         SELECT 1 FROM inserted
69         WHERE team1_id = team2_id
70     )
71     BEGIN
72         RAISERROR('A team cannot play against itself.', 16, 1);
73         ROLLBACK;
74     END
75 END;
76
77 -- team1_id = team2_id = 1 + trigger fires
78 INSERT INTO match VALUES (301, '2025-01-01', 'Lahore', 1, 1, NULL, 1);
79
80 -- Q5
81 CREATE TRIGGER trg_match_winner_valid
82 ON match
83 AFTER UPDATE
84 AS
85 BEGIN
```

110 % 2 0 11 Ln: 77, Ch: 1 (114 chars, 2 lines) SPC CRLF

Messages

Msg 50000, Level 16, State 1, Procedure trg\_match\_same\_team, Line 12 [Batch Start Line 76]  
A team cannot play against itself.

Msg 3609, Level 16, State 1, Line 78  
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-04-17T12:46:20.4857547-07:00

Q5

```

76
77 -- team1_id = team2_id = 1 → trigger fires
78 INSERT INTO match VALUES (301, '2025-01-01', 'Lahore', 1, 1, NULL, 1);
79
80 -- Q5
81 CREATE TRIGGER trg_match_winner_valid
82 ON match
83 AFTER UPDATE
84 AS
85 BEGIN
86     IF EXISTS (
87         SELECT 1 FROM inserted
88         WHERE winner_id IS NOT NULL
89             AND winner_id != team1_id
90             AND winner_id != team2_id
91     )
92     BEGIN
93         RAISERROR('Winner must be one of the two teams in the match.', 16, 1);
94         ROLLBACK;
95     END
96 END;
97

```

110 % 2 0 ↑ ↓ Ln: 96, Ch: 5 (371 chars, 17 lines) SPC CRLF UTF

Messages

Commands completed successfully.

Completion time: 2026-04-17T12:46:42.0528312-07:00

```

82  ON match
83  AFTER UPDATE
84  AS
85  BEGIN
86      IF EXISTS (
87          SELECT 1 FROM inserted
88          WHERE winner_id IS NOT NULL
89              AND winner_id != team1_id
90              AND winner_id != team2_id
91      )
92      BEGIN
93          RAISERROR('Winner must be one of the two teams in the match.', 16, 1);
94          ROLLBACK;
95      END
96  END;
97
98  -- match 201 is between team 1 and 2, setting winner to team 3 + trigger fires
99  UPDATE match SET winner_id = 3 WHERE mid = 201;
100
101  -- Q6
102  CREATE TRIGGER trg_performance_team_check
103  ON performance
104  AFTER UPDATE
105  AS
106  BEGIN
107      IF EXISTS (
108          SELECT 1 FROM inserted
109          WHERE team_id IS NOT NULL
110              AND team_id != team1_id
111              AND team_id != team2_id
112      )
113      BEGIN
114          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
115          ROLLBACK;
116      END
117  END;
118
119  -- Q7
120  CREATE TRIGGER trg_performance_team_check
121  ON performance
122  AFTER UPDATE
123  AS
124  BEGIN
125      IF EXISTS (
126          SELECT 1 FROM inserted
127          WHERE team_id IS NOT NULL
128              AND team_id != team1_id
129              AND team_id != team2_id
130      )
131      BEGIN
132          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
133          ROLLBACK;
134      END
135  END;
136
137  -- Q8
138  CREATE TRIGGER trg_performance_team_check
139  ON performance
140  AFTER UPDATE
141  AS
142  BEGIN
143      IF EXISTS (
144          SELECT 1 FROM inserted
145          WHERE team_id IS NOT NULL
146              AND team_id != team1_id
147              AND team_id != team2_id
148      )
149      BEGIN
150          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
151          ROLLBACK;
152      END
153  END;
154
155  -- Q9
156  CREATE TRIGGER trg_performance_team_check
157  ON performance
158  AFTER UPDATE
159  AS
160  BEGIN
161      IF EXISTS (
162          SELECT 1 FROM inserted
163          WHERE team_id IS NOT NULL
164              AND team_id != team1_id
165              AND team_id != team2_id
166      )
167      BEGIN
168          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
169          ROLLBACK;
170      END
171  END;
172
173  -- Q10
174  CREATE TRIGGER trg_performance_team_check
175  ON performance
176  AFTER UPDATE
177  AS
178  BEGIN
179      IF EXISTS (
180          SELECT 1 FROM inserted
181          WHERE team_id IS NOT NULL
182              AND team_id != team1_id
183              AND team_id != team2_id
184      )
185      BEGIN
186          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
187          ROLLBACK;
188      END
189  END;
190
191  -- Q11
192  CREATE TRIGGER trg_performance_team_check
193  ON performance
194  AFTER UPDATE
195  AS
196  BEGIN
197      IF EXISTS (
198          SELECT 1 FROM inserted
199          WHERE team_id IS NOT NULL
200              AND team_id != team1_id
201              AND team_id != team2_id
202      )
203      BEGIN
204          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
205          ROLLBACK;
206      END
207  END;
208
209  -- Q12
210  CREATE TRIGGER trg_performance_team_check
211  ON performance
212  AFTER UPDATE
213  AS
214  BEGIN
215      IF EXISTS (
216          SELECT 1 FROM inserted
217          WHERE team_id IS NOT NULL
218              AND team_id != team1_id
219              AND team_id != team2_id
220      )
221      BEGIN
222          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
223          ROLLBACK;
224      END
225  END;
226
227  -- Q13
228  CREATE TRIGGER trg_performance_team_check
229  ON performance
230  AFTER UPDATE
231  AS
232  BEGIN
233      IF EXISTS (
234          SELECT 1 FROM inserted
235          WHERE team_id IS NOT NULL
236              AND team_id != team1_id
237              AND team_id != team2_id
238      )
239      BEGIN
240          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
241          ROLLBACK;
242      END
243  END;
244
245  -- Q14
246  CREATE TRIGGER trg_performance_team_check
247  ON performance
248  AFTER UPDATE
249  AS
250  BEGIN
251      IF EXISTS (
252          SELECT 1 FROM inserted
253          WHERE team_id IS NOT NULL
254              AND team_id != team1_id
255              AND team_id != team2_id
256      )
257      BEGIN
258          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
259          ROLLBACK;
260      END
261  END;
262
263  -- Q15
264  CREATE TRIGGER trg_performance_team_check
265  ON performance
266  AFTER UPDATE
267  AS
268  BEGIN
269      IF EXISTS (
270          SELECT 1 FROM inserted
271          WHERE team_id IS NOT NULL
272              AND team_id != team1_id
273              AND team_id != team2_id
274      )
275      BEGIN
276          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
277          ROLLBACK;
278      END
279  END;
280
281  -- Q16
282  CREATE TRIGGER trg_performance_team_check
283  ON performance
284  AFTER UPDATE
285  AS
286  BEGIN
287      IF EXISTS (
288          SELECT 1 FROM inserted
289          WHERE team_id IS NOT NULL
290              AND team_id != team1_id
291              AND team_id != team2_id
292      )
293      BEGIN
294          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
295          ROLLBACK;
296      END
297  END;
298
299  -- Q17
300  CREATE TRIGGER trg_performance_team_check
301  ON performance
302  AFTER UPDATE
303  AS
304  BEGIN
305      IF EXISTS (
306          SELECT 1 FROM inserted
307          WHERE team_id IS NOT NULL
308              AND team_id != team1_id
309              AND team_id != team2_id
310      )
311      BEGIN
312          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
313          ROLLBACK;
314      END
315  END;
316
317  -- Q18
318  CREATE TRIGGER trg_performance_team_check
319  ON performance
320  AFTER UPDATE
321  AS
322  BEGIN
323      IF EXISTS (
324          SELECT 1 FROM inserted
325          WHERE team_id IS NOT NULL
326              AND team_id != team1_id
327              AND team_id != team2_id
328      )
329      BEGIN
330          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
331          ROLLBACK;
332      END
333  END;
334
335  -- Q19
336  CREATE TRIGGER trg_performance_team_check
337  ON performance
338  AFTER UPDATE
339  AS
340  BEGIN
341      IF EXISTS (
342          SELECT 1 FROM inserted
343          WHERE team_id IS NOT NULL
344              AND team_id != team1_id
345              AND team_id != team2_id
346      )
347      BEGIN
348          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
349          ROLLBACK;
350      END
351  END;
352
353  -- Q20
354  CREATE TRIGGER trg_performance_team_check
355  ON performance
356  AFTER UPDATE
357  AS
358  BEGIN
359      IF EXISTS (
360          SELECT 1 FROM inserted
361          WHERE team_id IS NOT NULL
362              AND team_id != team1_id
363              AND team_id != team2_id
364      )
365      BEGIN
366          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
367          ROLLBACK;
368      END
369  END;
370
371  -- Q21
372  CREATE TRIGGER trg_performance_team_check
373  ON performance
374  AFTER UPDATE
375  AS
376  BEGIN
377      IF EXISTS (
378          SELECT 1 FROM inserted
379          WHERE team_id IS NOT NULL
380              AND team_id != team1_id
381              AND team_id != team2_id
382      )
383      BEGIN
384          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
385          ROLLBACK;
386      END
387  END;
388
389  -- Q22
390  CREATE TRIGGER trg_performance_team_check
391  ON performance
392  AFTER UPDATE
393  AS
394  BEGIN
395      IF EXISTS (
396          SELECT 1 FROM inserted
397          WHERE team_id IS NOT NULL
398              AND team_id != team1_id
399              AND team_id != team2_id
400      )
401      BEGIN
402          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
403          ROLLBACK;
404      END
405  END;
406
407  -- Q23
408  CREATE TRIGGER trg_performance_team_check
409  ON performance
410  AFTER UPDATE
411  AS
412  BEGIN
413      IF EXISTS (
414          SELECT 1 FROM inserted
415          WHERE team_id IS NOT NULL
416              AND team_id != team1_id
417              AND team_id != team2_id
418      )
419      BEGIN
420          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
421          ROLLBACK;
422      END
423  END;
424
425  -- Q24
426  CREATE TRIGGER trg_performance_team_check
427  ON performance
428  AFTER UPDATE
429  AS
430  BEGIN
431      IF EXISTS (
432          SELECT 1 FROM inserted
433          WHERE team_id IS NOT NULL
434              AND team_id != team1_id
435              AND team_id != team2_id
436      )
437      BEGIN
438          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
439          ROLLBACK;
440      END
441  END;
442
443  -- Q25
444  CREATE TRIGGER trg_performance_team_check
445  ON performance
446  AFTER UPDATE
447  AS
448  BEGIN
449      IF EXISTS (
450          SELECT 1 FROM inserted
451          WHERE team_id IS NOT NULL
452              AND team_id != team1_id
453              AND team_id != team2_id
454      )
455      BEGIN
456          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
457          ROLLBACK;
458      END
459  END;
460
461  -- Q26
462  CREATE TRIGGER trg_performance_team_check
463  ON performance
464  AFTER UPDATE
465  AS
466  BEGIN
467      IF EXISTS (
468          SELECT 1 FROM inserted
469          WHERE team_id IS NOT NULL
470              AND team_id != team1_id
471              AND team_id != team2_id
472      )
473      BEGIN
474          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
475          ROLLBACK;
476      END
477  END;
478
479  -- Q27
480  CREATE TRIGGER trg_performance_team_check
481  ON performance
482  AFTER UPDATE
483  AS
484  BEGIN
485      IF EXISTS (
486          SELECT 1 FROM inserted
487          WHERE team_id IS NOT NULL
488              AND team_id != team1_id
489              AND team_id != team2_id
490      )
491      BEGIN
492          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
493          ROLLBACK;
494      END
495  END;
496
497  -- Q28
498  CREATE TRIGGER trg_performance_team_check
499  ON performance
500  AFTER UPDATE
501  AS
502  BEGIN
503      IF EXISTS (
504          SELECT 1 FROM inserted
505          WHERE team_id IS NOT NULL
506              AND team_id != team1_id
507              AND team_id != team2_id
508      )
509      BEGIN
510          RAISERROR('Team must be one of the two teams in the match.', 16, 1);
511          ROLLBACK;
512      END
513  END;
514
515  -- Q29
516  CREATE TRIGGER trg_performance_team_check
517  ON performance
518  AFTER UPDATE
519  AS
520  BEGIN
521      IF EXISTS (
522          SELECT 1 FROM inserted
523          WHERE team_id IS NOT NULL
524              AND team_id != team1_id
525              AND team_id != team2
```

Q6

```

97
98 -- match 201 is between team 1 and 2, setting winner to team 3 → trigger fires
99 UPDATE match SET winner_id = 3 WHERE mid = 201;
100
101 -- Q6
102 CREATE TRIGGER trg_performance_team_check
103 ON performance
104 AFTER INSERT
105 AS
106 BEGIN
107     IF EXISTS (
108         SELECT 1 FROM inserted i
109         JOIN player p ON i.pid = p.pid
110         JOIN match m ON i.mid = m.mid
111         WHERE p.tid != m.team1_id
112             AND p.tid != m.team2_id
113     )
114     BEGIN
115         RAISERROR('Player team did not participate in this match.', 16, 1);
116     ROLLBACK;
117     END
118 END;

```

110 % ▾ ● 2 ▲ 0 ↑ ▾ Ln: 118, Ch: 5 (419 chars, 18 lines) SPC CRLF UT

Messages

Commands completed successfully.

Completion time: 2026-04-17T12:47:15.9131256-07:00

```
241-0583 Lab...d Raza (78)  ⌵  ⌵  ⌵  ...
100
101 -- Q6
102 CREATE TRIGGER trg_performance_team_check
103 ON performance
104 AFTER INSERT
105 AS
106 BEGIN
107     IF EXISTS (
108         SELECT 1 FROM inserted i
109         JOIN player p ON i.pid = p.pid
110         JOIN match m ON i.mid = m.mid
111         WHERE p.tid != m.team1_id
112             AND p.tid != m.team2_id
113     )
114     BEGIN
115         RAISERROR('Player team did not participate in this match.', 16, 1);
116         ROLLBACK;
117     END
118 END;
119
120 -- pid 107 belongs to team 4, match 201 is between team 1 and 2 → trigger fires
121 INSERT INTO performance VALUES (107, 201, 30, 1, 0);
122
110 % ⌵  ●  ⚠  0  ⬆  ⬇
Messages
(0 rows affected)

(1 row affected)
Total rows added in this operation: 30
Msg 50000, Level 16, State 1, Procedure trg_performance_team_check, Line 15 [Batch Start Line 119]
Player team did not participate in this match.
Msg 3609, Level 16, State 1, Line 121
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-04-17T12:47:25.4973738-07:00
```

Q7

```
24L-0583 Lab...d Raza (78) X
118 END;
119
120 -- pid 107 belongs to team 4, match 201 is between team 1 and 2 + trigger fires
121 INSERT INTO performance VALUES (107, 201, 30, 1, 0);
122
123 -- Q7
124 CREATE TRIGGER trg_performance_not_both_zero
125 ON performance
126 AFTER UPDATE
127 AS
128 BEGIN
129     IF EXISTS (
130         SELECT 1 FROM inserted
131         WHERE runs = 0 AND wickets = 0
132     )
133     BEGIN
134         RAISERROR('Runs and wickets cannot both be zero.', 16, 1);
135         ROLLBACK;
136     END
137 END;
138
139 -- pid 101, mid 201 currently has runs=75, setting both to 0 + trigger fires
140
```

Messages

Commands completed successfully.

Completion time: 2026-04-17T12:45:28.2732239-07:00

```
24L-0583 Lab...d Raza (78) X
121 INSERT INTO performance VALUES (107, 201, 30, 1, 0);
122
123 -- Q7
124 CREATE TRIGGER trg_performance_not_both_zero
125 ON performance
126 AFTER UPDATE
127 AS
128 BEGIN
129     IF EXISTS (
130         SELECT 1 FROM inserted
131         WHERE runs = 0 AND wickets = 0
132     )
133     BEGIN
134         RAISERROR('Runs and wickets cannot both be zero.', 16, 1);
135         ROLLBACK;
136     END
137 END;
138
139 -- pid 101, mid 201 currently has runs=75, setting both to 0 + trigger fires
140 UPDATE performance SET runs = 0, wickets = 0 WHERE pid = 101 AND mid = 201;
141
142 -- Q8
```

Messages

Msg 50000, Level 16, State 1, Procedure trg\_performance\_not\_both\_zero, Line 12 [Batch Start Line 138]  
Runs and wickets cannot both be zero.  
Msg 3609, Level 16, State 1, Line 140  
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-04-17T12:48:40.3759623-07:00

Q8

```

136      END
137    END;
138
139    -- pid 101, mid 201 currently has runs=75, setting both to 0 + trigger fires
140    UPDATE performance SET runs = 0, wickets = 0 WHERE pid = 101 AND mid = 201;
141
142    -- Q8
143    CREATE TRIGGER trg_tournament_delete_check
144    ON tournament
145    AFTER DELETE
146    AS
147    BEGIN
148        IF EXISTS (
149            SELECT 1 FROM match m
150            JOIN deleted d ON m tour_id = d tour_id
151        )
152        BEGIN
153            RAISERROR('Cannot delete tournament: matches exist for it.', 16, 1);
154            ROLLBACK;
155        END
156    END;
157

```

Ln: 156, Ch: 5 (316 chars, 15 lines) SPC CRLF UTF-8

Messages

Commands completed successfully.

Completion time: 2026-04-17T12:48:52.8186567-07:00

```

241-0583 Lab...d Raza (78)
139 -- pid 101, mid 201 currently has runs=75, setting both to 0 → trigger fires
140 UPDATE performance SET runs = 0, wickets = 0 WHERE pid = 101 AND mid = 201;
141
142 -- Q8
143 CREATE TRIGGER trg_tournament_delete_check
144 ON tournament
145 AFTER DELETE
146 AS
147 BEGIN
148     IF EXISTS (
149         SELECT 1 FROM match m
150         JOIN deleted d ON m.tour_id = d.tour_id
151     )
152     BEGIN
153         RAISERROR('Cannot delete tournament: matches exist for it.', 16, 1);
154         ROLLBACK;
155     END
156 END;
157
158 -- tour_id 1 has matches 201, 202, 203 linked to it → trigger fires
159 DELETE FROM tournament WHERE tour_id = 1;
160
110 % 2 0 ↑ ↓ Ln: 158, Ch: 1 (110 chars, 2 lines) SPC CRLF
Messages
Msg 547, Level 16, State 0, Line 159
The DELETE statement conflicted with the REFERENCE constraint "FK_match_tour_id_5535A963". The conflict occurred in database "cricket", table "dbo.match", column 'tour_id'.
The statement has been terminated.
Completion time: 2026-04-17T12:49:00.5253759-07:00

```

Q9

```
241-0583 Lab...d Raza (78) X
157
158 -- tour_id 1 has matches 201, 202, 203 linked to it + trigger fires
159 DELETE FROM tournament WHERE tour_id = 1;
160
161 -- Q9
162 CREATE TRIGGER trg_match_min_players
163 ON match
164 AFTER INSERT
165 AS
166 BEGIN
167     IF EXISTS (
168         SELECT 1 FROM inserted i
169         WHERE (SELECT COUNT(*) FROM player WHERE tid = i.team1_id) < 11
170              OR (SELECT COUNT(*) FROM player WHERE tid = i.team2_id) < 11
171     )
172     BEGIN
173         RAISERROR('Both teams must have at least 11 players.', 16, 1);
174         ROLLBACK;
175     END
176 END;
177
178 -- team 1 has only 2 players, team 2 has only 2, both < 11 + trigger fires
179
110% 2 0 10% Ln: 161, Ch: 1 (399 chars, 16 lines) SPC CRLF U
Messages
Commands completed successfully.
Completion time: 2026-04-17T12:49:19.3872335-07:00
```

```
241-0583 Lab...d Raza (78) X
160
161 -- Q9
162 CREATE TRIGGER trg_match_min_players
163 ON match
164 AFTER INSERT
165 AS
166 BEGIN
167     IF EXISTS (
168         SELECT 1 FROM inserted i
169         WHERE (SELECT COUNT(*) FROM player WHERE tid = i.team1_id) < 11
170              OR (SELECT COUNT(*) FROM player WHERE tid = i.team2_id) < 11
171     )
172     BEGIN
173         RAISERROR('Both teams must have at least 11 players.', 16, 1);
174         ROLLBACK;
175     END
176 END;
177
178 -- team 1 has only 2 players, team 2 has only 2, both < 11 + trigger fires
179 INSERT INTO match VALUES (302, '2025-05-01', 'Karachi', 1, 2, NULL, 2);
180
181 -- Q10
182
110% 2 0 10% Ln: 178, Ch: 1 (147 chars, 2 lines) SPC CRLF UTF-
Messages
Msg 50000, Level 16, State 1, Procedure trg_match_min_players, Line 13 [Batch Start Line 177]
Both teams must have at least 11 players.
Msg 3609, Level 16, State 1, Line 179
The transaction ended in the trigger. The batch has been aborted.
Completion time: 2026-04-17T12:49:28.0926439-07:00
```

## Q10

```

241-0583 Lab...d Raza (78)
178 -- team 1 has only 2 players, team 2 has only 2, both < 11 + trigger fires
179 INSERT INTO match VALUES (302, '2025-05-01', 'Karachi', 1, 2, NULL, 2);
180
181 -- Q10
182 CREATE TRIGGER trg_player_salary_cap
183 ON player
184 AFTER UPDATE
185 AS
186 BEGIN
187     IF EXISTS (
188         SELECT 1 FROM inserted i
189         WHERE i.salary > 2 * (
190             SELECT AVG(salary) FROM player WHERE tid = i.tid
191         )
192     )
193     BEGIN
194         RAISERROR('Salary exceeds twice the team average salary.', 16, 1);
195         ROLLBACK;
196     END
197 END;
198
199 -- team 1 avg salary = 87500, twice = 175000, setting 999999 + trigger fires
200

```

Ln: 181, Ch: 1 | (364 chars, 17 lines) | SPC CRLF UT

Messages

Commands completed successfully.

Completion time: 2026-04-17T12:49:45.9135753-07:00

```

241-0583 Lab...d Raza (78)
184 AFTER UPDATE
185 AS
186 BEGIN
187     IF EXISTS (
188         SELECT 1 FROM inserted i
189         WHERE i.salary > 2 * (
190             SELECT AVG(salary) FROM player WHERE tid = i.tid
191         )
192     )
193     BEGIN
194         RAISERROR('Salary exceeds twice the team average salary.', 16, 1);
195         ROLLBACK;
196     END
197 END;
198
199 -- team 1 avg salary = 87500, twice = 175000, setting 999999 + trigger fires
200 UPDATE player SET salary = 999999 WHERE pid = 101;

```

Ln: 199, Ch: 1 | (128 chars, 2 lines) | SPC CRLF U

Messages

Salary changed by more than 5000.

(1 row affected)

Msg 50000, Level 16, State 1, Procedure trg\_player\_salary\_cap, Line 14 [Batch Start Line 198]  
 Salary exceeds twice the team average salary.  
 Msg 3609, Level 16, State 1, Line 200  
 The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-04-17T12:49:58.2373745-07:00